MAT2002 HW6 Solutions

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Here are the rigorous mathematical step-by-step solutions for the problems presented in your assignment.

1. The Gamma Function

The gamma function is defined by the integral $\Gamma(p+1) = \int_0^\infty e^{-x}x^p dx$.

(a) Show that, for $p > 0$, $\Gamma(p+1) = p\Gamma(p)$

We use integration by parts. Let $u = x^p$ and $dv = e^{-x}dx$. This gives $du = px^{p-1}dx$ and $v = -e^{-x}$. Applying the integration by parts formula $\int u\,dv = uv - \int v\,du$:

$$\Gamma(p+1) = \left[-x^p e^{-x}\right]_0^\infty - \int_0^\infty (-e^{-x})(px^{p-1})dx \tag{1}$$

Since $p > 0$, $\lim_{x\to 0} x^p e^{-x} = 0$. By L'Hôpital's rule, polynomials grow slower than exponentials, so $\lim_{x\to\infty} \frac{x^p}{e^x} = 0$. The boundary term vanishes.

$$\Gamma(p+1) = 0 + p\int_0^\infty x^{p-1}e^{-x}dx = p\Gamma(p) \tag{2}$$

(b) Show that $\Gamma(1) = 1$

By setting $p = 0$ in the integral definition:

$$\Gamma(1) = \int_0^\infty e^{-x}x^0 dx = \int_0^\infty e^{-x}dx = \left[-e^{-x}\right]_0^\infty = 0 - (-1) = 1 \tag{3}$$

(c) If p is a positive integer n, show that $\Gamma(n+1) = n!$

We proceed by mathematical induction.
Base case ($n = 1$): From part (b), $\Gamma(2) = 1 \cdot \Gamma(1) = 1 = 1!$. This holds true.
Inductive step: Assume $\Gamma(k+1) = k!$ for some integer $k \geq 1$. For $n = k+1$, we use the recurrence relation from part (a):

$$\Gamma((k+1)+1) = (k+1)\Gamma(k+1) = (k+1)(k!) = (k+1)! \tag{4}$$

By induction, the statement is true for all positive integers n.

(d) Show that, for $p > 0$, $p(p+1)\cdots(p+n-1) = \Gamma(p+n)/\Gamma(p)$

Using the relation from part (a) recursively:

$$\Gamma(p+n) = (p+n-1)\Gamma(p+n-1) \tag{5}$$

$$\Gamma(p+n) = (p+n-1)(p+n-2)\Gamma(p+n-2) \tag{6}$$

Continuing this expansion n times yields:

$$\Gamma(p+n) = (p+n-1)(p+n-2)\cdots(p+1)p\Gamma(p) \tag{7}$$

Dividing both sides by $\Gamma(p)$ directly proves the identity.

Find $\Gamma(3/2)$ and $\Gamma(11/2)$
Given $\Gamma(1/2) = \sqrt{\pi}$:

$$\Gamma(3/2) = \Gamma(1/2+1) = \frac{1}{2}\Gamma(1/2) = \frac{\sqrt{\pi}}{2} \tag{8}$$

$$\Gamma(11/2) = \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\Gamma(3/2) = \frac{945}{16}\cdot\frac{\sqrt{\pi}}{2} = \frac{945\sqrt{\pi}}{32} \tag{9}$$

2. Laplace transform of t^p

Consider $\mathcal{L}\{t^p\}$ for $p > -1$.

(a) Show that $\mathcal{L}\{t^p\} = \Gamma(p+1)/s^{p+1}$, $s > 0$

By definition, $\mathcal{L}\{t^p\} = \int_0^\infty e^{-st}t^p dt$. Let $x = st$, which implies $t = x/s$ and $dt = dx/s$. Because $s > 0$, the limits of integration remain 0 to ∞.

$$\mathcal{L}\{t^p\} = \int_0^\infty e^{-x}\left(\frac{x}{s}\right)^p \frac{dx}{s} = \frac{1}{s^{p+1}}\int_0^\infty e^{-x}x^p dx \tag{10}$$

By the definition of the Gamma function, the integral evaluates to $\Gamma(p+1)$, giving $\frac{\Gamma(p+1)}{s^{p+1}}$.

(b) Show that $\mathcal{L}\{t^n\} = n!/s^{n+1}$

Substitute $p = n$ into the result from (a). From 1(c), we proved $\Gamma(n+1) = n!$. Therefore,

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \tag{11}$$

(c) Show that $\mathcal{L}\{t^{-1/2}\} = \frac{2}{\sqrt{s}}\int_0^\infty e^{-x^2}dx$

From the Laplace definition with $p = -1/2$: $\mathcal{L}\{t^{-1/2}\} = \int_0^\infty e^{-st}t^{-1/2}dt$. Use the substitution $x^2 = st$. Then $2x\,dx = s\,dt$, meaning $dt = \frac{2x}{s}dx$. Also, $t^{-1/2} = \frac{\sqrt{s}}{x}$.

$$\int_0^\infty e^{-x^2}\left(\frac{\sqrt{s}}{x}\right)\left(\frac{2x}{s}\right)dx = \int_0^\infty e^{-x^2}\frac{2}{\sqrt{s}}dx = \frac{2}{\sqrt{s}}\int_0^\infty e^{-x^2}dx \tag{12}$$

(d) Show that $\mathcal{L}\{t^{1/2}\} = \sqrt{\pi}/(2s^{3/2})$

Using the given integral identity $\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$, the result from (c) becomes $\mathcal{L}\{t^{-1/2}\} = \sqrt{\pi/s}$. To find $\mathcal{L}\{t^{1/2}\}$, we can use the formula from (a):

$$\mathcal{L}\{t^{1/2}\} = \frac{\Gamma(3/2)}{s^{3/2}} \tag{13}$$

As found in 1(d), $\Gamma(3/2) = \frac{\sqrt{\pi}}{2}$. Substituting this in yields $\frac{\sqrt{\pi}}{2s^{3/2}}$.

3. Solve IVPs with Laplace Transformations

(i) $y'' + 2y' + 5y = 0$; $y(0) = 2$, $y'(0) = -1$

Taking the Laplace transform of both sides:

$$(s^2Y(s) - sy(0) - y'(0)) + 2(sY(s) - y(0)) + 5Y(s) = 0 \tag{14}$$

Substitute the initial conditions:

$$(s^2Y(s) - 2s + 1) + 2(sY(s) - 2) + 5Y(s) = 0 \tag{15}$$

$$Y(s)[s^2 + 2s + 5] - 2s - 3 = 0 \implies Y(s) = \frac{2s+3}{s^2+2s+5} \tag{16}$$

Complete the square in the denominator: $s^2 + 2s + 5 = (s+1)^2 + 4$.

$$Y(s) = \frac{2(s+1)+1}{(s+1)^2+4} = 2\left(\frac{s+1}{(s+1)^2+4}\right) + \frac{1}{2}\left(\frac{2}{(s+1)^2+4}\right) \tag{17}$$

Taking the inverse Laplace transform:

$$y(t) = 2e^{-t}\cos(2t) + \frac{1}{2}e^{-t}\sin(2t) \tag{18}$$

(ii) $y^{(4)} - 4y''' + 6y'' - 4y' + y = 0$; $y(0) = y''(0) = 0$, $y'(0) = y'''(0) = 1$

The polynomial factors as $(s-1)^4 = 0$. Transforming the equation:

$$(s^4Y - s^3(0) - s^2(1) - s(0) - 1) - 4(s^3Y - s^2(0) - s(1) - 0) + 6(s^2Y - 1) - 4(sY) + Y = 0 \tag{19}$$

$$Y(s)(s^4 - 4s^3 + 6s^2 - 4s + 1) - s^2 - 1 + 4s - 6 = 0 \tag{20}$$

$$(s-1)^4Y(s) = s^2 - 4s + 7 \tag{21}$$

Express the numerator in terms of $(s-1)$: $s^2 - 4s + 7 = (s-1)^2 - 2(s-1) + 4$.

$$Y(s) = \frac{(s-1)^2 - 2(s-1) + 4}{(s-1)^4} = \frac{1}{(s-1)^2} - \frac{2}{(s-1)^3} + \frac{4}{(s-1)^4} \tag{22}$$

Using $\mathcal{L}^{-1}\left\{\frac{n!}{(s-a)^{n+1}}\right\} = t^n e^{at}$:

$$y(t) = te^t - t^2e^t + \frac{2}{3}t^3e^t \tag{23}$$

4. Laplace Transformation of a Piecewise Function

Find the transform for $f(t) = t - \pi$ for $\pi \leq t < 2\pi$, and 0 otherwise.
Express $f(t)$ using Heaviside step functions $u_c(t)$:

$$f(t) = (t-\pi)[u_\pi(t) - u_{2\pi}(t)] = (t-\pi)u_\pi(t) - (t-\pi)u_{2\pi}(t) \tag{24}$$

To transform $(t-\pi)u_{2\pi}(t)$, rewrite $(t-\pi)$ as $(t-2\pi)+\pi$:

$$f(t) = (t-\pi)u_\pi(t) - (t-2\pi)u_{2\pi}(t) - \pi u_{2\pi}(t) \tag{25}$$

Using the property $\mathcal{L}\{f(t-c)u_c(t)\} = e^{-cs}F(s)$:

$$\mathcal{L}\{f(t)\} = \frac{e^{-\pi s}}{s^2} - \frac{e^{-2\pi s}}{s^2} - \frac{\pi e^{-2\pi s}}{s} \tag{26}$$

5. Solve the Given Initial Value Problems

(i) $y'' + y' + \frac{5}{4}y = g(t)$; $y(0) = y'(0) = 0$, $g(t) = \sin t$ for $0 \leq t < \pi$, 0 elsewhere

Rewrite $g(t) = \sin(t) - \sin(t)u_\pi(t) = \sin(t) + \sin(t-\pi)u_\pi(t)$.

$$G(s) = \frac{1}{s^2+1} + \frac{e^{-\pi s}}{s^2+1} \tag{27}$$

Transforming the ODE: $(s^2 + s + \frac{5}{4})Y(s) = G(s)$. Let $H(s) = \frac{1}{(s^2+s+5/4)(s^2+1)}$.
Using partial fractions: $\frac{1}{(s^2+s+5/4)(s^2+1)} = \frac{As+B}{s^2+s+5/4} + \frac{Cs+D}{s^2+1}$.
Solving yields $A = 16/17, B = 12/17, C = -16/17, D = 4/17$.

$$H(s) = \frac{16}{17}\left(\frac{s+1/2}{(s+1/2)^2+1}\right) + \frac{4}{17}\left(\frac{1}{(s+1/2)^2+1}\right) - \frac{16}{17}\left(\frac{s}{s^2+1}\right) + \frac{4}{17}\left(\frac{1}{s^2+1}\right) \tag{28}$$

Inverse transforming $H(s)$ to find $h(t)$:

$$h(t) = \frac{16}{17}e^{-t/2}\cos(t) + \frac{4}{17}e^{-t/2}\sin(t) - \frac{16}{17}\cos(t) + \frac{4}{17}\sin(t) \tag{29}$$

Since $Y(s) = H(s) + e^{-\pi s}H(s)$, the final solution is:

$$y(t) = h(t) + h(t-\pi)u_\pi(t) \tag{30}$$

(ii) $y^{(4)} - y = u_1(t) - u_2(t)$; zero initial conditions

$$(s^4-1)Y(s) = \frac{e^{-s}}{s} - \frac{e^{-2s}}{s} \implies Y(s) = \frac{1}{s(s^4-1)}(e^{-s} - e^{-2s}) \tag{31}$$

Let $H(s) = \frac{1}{s(s^4-1)} = \frac{1}{s(s-1)(s+1)(s^2+1)}$. Using partial fractions:

$$H(s) = -\frac{1}{s} + \frac{1/4}{s-1} + \frac{1/4}{s+1} + \frac{1/2s}{s^2+1} \tag{32}$$

Taking the inverse transform:

$$h(t) = -1 + \frac{1}{4}e^t + \frac{1}{4}e^{-t} + \frac{1}{2}\cos(t) = -1 + \frac{1}{2}\cosh(t) + \frac{1}{2}\cos(t) \tag{33}$$

By the second shifting theorem:

$$y(t) = h(t-1)u_1(t) - h(t-2)u_2(t) \tag{34}$$

6. Delta Function Initial Value Problems

(i) $y'' - y = -20\delta(t-3)$; $y(0) = 1$, $y'(0) = 0$

$$(s^2Y - s) - Y = -20e^{-3s} \implies Y(s)(s^2-1) = s - 20e^{-3s} \tag{35}$$

$$Y(s) = \frac{s}{s^2-1} - 20\frac{e^{-3s}}{s^2-1} \tag{36}$$

Taking the inverse Laplace transform:

$$y(t) = \cosh(t) - 20\sinh(t-3)u_3(t) \tag{37}$$

(ii) $y'' + 2y' + 2y = \cos t + \delta(t-\pi/2)$; $y(0) = 0, y'(0) = 0$

$$(s^2+2s+2)Y(s) = \frac{s}{s^2+1} + e^{-\pi s/2} \tag{38}$$

$$Y(s) = \frac{s}{(s^2+1)(s^2+2s+2)} + \frac{e^{-\pi s/2}}{s^2+2s+2} = H_1(s) + H_2(s)e^{-\pi s/2} \tag{39}$$

For $H_1(s)$, partial fractions give:

$$H_1(s) = \frac{1}{5}\left(\frac{s+2}{s^2+1}\right) - \frac{1}{5}\left(\frac{s+4}{(s+1)^2+1}\right) \tag{40}$$

$$h_1(t) = \frac{1}{5}\cos(t) + \frac{2}{5}\sin(t) - \frac{1}{5}e^{-t}\cos(t) - \frac{3}{5}e^{-t}\sin(t) \tag{41}$$

For $H_2(s) = \frac{1}{(s+1)^2+1}$, we have $h_2(t) = e^{-t}\sin(t)$.

$$y(t) = h_1(t) + h_2(t-\pi/2)u_{\pi/2}(t) \tag{42}$$

Using $\sin(t-\pi/2) = -\cos(t)$, the shift expands to:

$$y(t) = h_1(t) - e^{-(t-\pi/2)}\cos(t)u_{\pi/2}(t) \tag{43}$$

7. General Volterra Integral Equation

Given $\phi(t) + \int_0^t k(t-\xi)\phi(\xi)d\xi = f(t)$, we must find $\mathcal{L}\{\phi(t)\}$.
The integral is a convolution $(k * \phi)(t)$. By the Convolution Theorem, $\mathcal{L}\{(k*\phi)(t)\} = K(s)\Phi(s)$.

$$\Phi(s) + K(s)\Phi(s) = F(s) \tag{44}$$

$$\Phi(s)(1+K(s)) = F(s) \implies \Phi(s) = \frac{F(s)}{1+K(s)} \tag{45}$$

8. Solving a Specific Volterra Equation

Consider $\phi(t) + \int_0^t (t-\xi)\phi(\xi)d\xi = \sin(2t)$.

(a) Solve the integral equation by using the Laplace transform

Here $k(t) = t \implies K(s) = 1/s^2$ and $f(t) = \sin(2t) \implies F(s) = 2/(s^2+4)$. Applying the formula from Q7:

$$\Phi(s) = \frac{2/(s^2+4)}{1+1/s^2} = \frac{2s^2}{(s^2+1)(s^2+4)} \tag{46}$$

Using partial fractions on $u = s^2$: $\frac{2u}{(u+1)(u+4)} = \frac{-2/3}{u+1} + \frac{8/3}{u+4}$.

$$\Phi(s) = -\frac{2}{3}\left(\frac{1}{s^2+1}\right) + \frac{4}{3}\left(\frac{2}{s^2+4}\right) \tag{47}$$

Inverse transforming yields:

$$\phi(t) = -\frac{2}{3}\sin(t) + \frac{4}{3}\sin(2t) \tag{48}$$

(b) Show $\phi(t)$ satisfies $\phi''(t)+\phi(t) = -4\sin(2t)$ and $\phi(0) = 0$, $\phi'(0) = 2$

We differentiate the integral equation $\phi(t) + \int_0^t (t-\xi)\phi(\xi)d\xi = \sin(2t)$ twice. Applying the Leibniz integral rule for the first derivative:

$$\phi'(t) + \int_0^t \phi(\xi)d\xi + (t-t)\phi(t) = 2\cos(2t) \implies \phi'(t) + \int_0^t \phi(\xi)d\xi = 2\cos(2t) \tag{49}$$

Differentiating a second time:

$$\phi''(t) + \phi(t) = -4\sin(2t) \tag{50}$$

For initial conditions, evaluate the equations at $t = 0$:

$$\phi(0) + \int_0^0 (0-\xi)\phi(\xi)d\xi = \sin(0) \implies \phi(0) = 0 \tag{51}$$

$$\phi'(0) + \int_0^0 \phi(\xi)d\xi = 2\cos(0) \implies \phi'(0) = 2 \tag{52}$$

(c) Solve the IVP in part (b), and verify the solution matches part (a)

The characteristic equation for the homogeneous part $r^2 + 1 = 0$ gives roots $\pm i$, so $\phi_h(t) = c_1\cos(t) + c_2\sin(t)$.
For the particular solution, guess $\phi_p(t) = A\cos(2t) + B\sin(2t)$.
Substituting into $\phi'' + \phi = -4\sin(2t)$ yields $-3A\cos(2t) - 3B\sin(2t) = -4\sin(2t)$, meaning $A = 0$ and $B = 4/3$.
The general solution is $\phi(t) = c_1\cos(t) + c_2\sin(t) + \frac{4}{3}\sin(2t)$.
Applying initial conditions:

$$\phi(0) = c_1 = 0 \tag{53}$$

$$\phi'(t) = c_2\cos(t) + \frac{8}{3}\cos(2t) \implies \phi'(0) = c_2 + \frac{8}{3} = 2 \implies c_2 = -\frac{2}{3} \tag{54}$$

Thus,

$$\phi(t) = -\frac{2}{3}\sin(t) + \frac{4}{3}\sin(2t) \tag{55}$$

This exactly matches the result found in part (a).